Yujie Li PhD Marie Skłodowska-Curie Actions Fellow

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Corresponding Member TC 209 ISSMGE Offshore Geotechnics

Academic links：https://scholar.google.com.hk/citations?user=RSLaP84AAAAJ&hl=zh-CN



Higher Education/ Degree

2018-2023	**Zhejiang University, Hangzhou, China** *Ph.D.* in Ocean Technology and Engineering Major: Ocean Engineering Minor: Geotechnical Engineering Dissertation "Microbial reinforcement technology for soils" Research Advisors: Professors Lizhong Wang and Zhen Guo
2014-2018	**Wuhan University of Science and Technology, Wuhan, China** Bachelor of Civil Engineering

Employment

2026-Present	**Marie Skłodowska-Curie Actions Fellow** Karlsruhe Institute of Technology (KIT), Germany, Host: Prof. Hans Henning Stutz Research: Soil-structure interfaces under microbial stimulation
2025-2026	**Research Fellow** Nanyang Technological University, Research Advisor: Professor Jian Chu Research: Nature-based coastal protection solutions
2024-2025	**Postdoctoral Research Fellow** Zhejiang University, Research Advisors: Professors Lizhong Wang and Daoqiong Zheng Research: Offshore geotechnical materials and sensing technology

Research Interests

Marine biogeotechnics	Biomineralization, MICP/EICP, multiscale experiments, coupled numerical model
Interface mechanics	Soil-structure interaction, constitutive model, bio-cemented interfaces
Marine geo-energy	Nature-based solutions for infrastructure protection: erosion, liquefaction and subsidence control, natural gas hydrate production, coupled THMC model
Subsea carbon storage	Bio-mediated sealing, reactive transport, storage integrity, leakage-risk control

Academic skills

Computational Tools	COMSOL (Advanced), MATLAB (Advanced), Abaqus/Plaxis (Proficient)
Experimental Skills	1 g and high g physical modeling, Field monitoring and testing

Competitive Research Funding as Principal Investigator

2026-2028	Marie Skłodowska-Curie Actions Postdoctoral Fellowships (~€202,125.12, PI, Grant No. 101270525), Interface Shear Behavior between Soil and Microbial-Stimulated Piles
2025-present	National Natural Science Foundation of China (~CNY 300 k, PI, Grant No. 52401344), Study on the MICP reaction-additional stress coupling of in-situ reinforcement of sand.

2024-2025	Nationally Funded Postdoctoral Fellowship (~CNY 120 k, PI, Grant No. GZC20241516), Marine soil and monopile strengthening by MICP.
2025-2025	The 77th batch of postdoctoral science funding (~CNY 80 k, PI, Grant No. 2025M773149), Rapid evaluation of microbial reinforcement of sandy soil.
2022-2023	Sanya Yazhou Bay Science and Technology City Administration doctoral funding (~CNY 100k, PI, Grant No. HSPHDSRF-2022-04-002), Ecological protection technology and reinforcement strategy for coastal erosion under extreme sea conditions.

Selected Collaborative Projects

2023-2026	Coastal Protection and Flood Resilience Institute (CFI)-Singapore (~$ 400 k, PI Jian Chu, Collaborator, Grant No.V1-P4), Eco-cement enhanced methods for beach erosion control and beach land restoration through soil accumulation
2021-2025	Ministry of National Development (MND)-Singapore (~$ 2,414 k, PI Jian Chu, Collaborator, Grant No.COT-V1-2020-4), Pilot field trials on the applications of bio-cementation for ground improvement works.
2025-2028	National Natural Science Foundation of China Key Project (~CNY 2,670 k, PI Lizhong Wang, Collaborator, Grant No. U24B20113), Mechanism and prevention of coupled buckling and axial displacement disasters in deep-sea pipelines under cyclic temperature and pressure.
2023-2027	National Natural Science Foundation of China Key Project (~CNY 2,530 k, PI Lizhong Wang, Collaborator, Grant No. 52238008), Dynamic disaster mechanism and design theory of wind power structure in typhoon-prone areas.
2021-2023	Hainan Major Science and Technology Projects (~CNY 12,000 k, PI Lizhong Wang, Collaborator, Grant No. ZDKJ202019), Multifactorial Disaster Mechanisms and Monitoring and Early Warning Technologies for Submarine Landslides.

Awards/Honors

2026	Marie Skłodowska-Curie Actions Postdoctoral Fellowships
2026	Zhejiang Provincial Outstanding Doctoral Dissertation Nomination Award
2025	China Communications and Transportation Association Science and Technology Progress Award, Ranking 5/15
2025	China Hydraulic Rock Mechanics Conference Outstanding Doctoral Dissertation Nomination Award
2024	Zhejiang University Outstanding Doctoral Dissertation Award
2024	Nationally Funded Postdoctoral Fellow
2024	China Electric Power Construction Science and Technology Progress Award, Ranking 15/15
2023	Graduate of Merit/Triple A graduate of Zhejiang University
2022	Graduate of Merit/Triple A graduate of Zhejiang University
2020	Graduate of Merit/Triple A graduate of Zhejiang University
2020	National Scholarship of China
2019	Graduate of Merit/Triple A graduate of Zhejiang University
2019	National Scholarship of China
2018	Outstanding Graduates of Wuhan University of Science and Technology
2016	National Encouragement Scholarship of China
2015	National Encouragement Scholarship of China

Professional Affiliations

2024-2028	Member of the International Society of Soil Mechanics and Geotechnical Engineering
2024-2026	Member of the Youth Committee of Environmental Geotechnical Engineering Branch of Chinese Society of Rock Mechanics and Engineering
2024-2028	Member of the Chinese Society of Rock Mechanics and Engineering
2024-2025	Member of the International Society for Rock Mechanics and Rock Engineering
2024-2025	Member of the Chinese Society of Mechanics
2024-2025	Member of the Chinese Geophysical Society

Editor

2026- present	Young editorial board member of Journal *Carbon Neutrality* (ESCI)
2026-present	Guest Editor of Special Issue *"Advanced Structure Analysis and Structure–Soil Interaction for Marine and Offshore Structures"* of *Journal of Marine Science and Engineering* (SCI)
2026-present	Young editorial board member of Journal *Materials Reports* (EI)
2026-present	Young editorial board member of Journal *Gas Science and Engineering* (SCI)
2026-present	Young editorial board member of Journal *Ocean*
2026-present	Young editorial board member of *Journal of Geomechanics and Measurements*
2025-present	Young editorial board member of Journal *Petroleum Science* (SCI)
2025-present	Guest Editor of Special Issue *"Advanced Technologies and Applications in Geotechnical Engineering" of Applied Sciences* (SCI)
2025-present	Young editorial board member of Journal *Rock Mechanics Bulletin* (ESCI)
2025-present	Young editorial board member of Journal *Smart Underground Engineering*
2024-2025	Young editorial board member of Journal *Biogeotechnics* (ESCI)
2024-present	Young editorial board member of Journal *Intelligent Geoengineering*

Research Supervisions

Ph.D.

Zhejiang University

1. Chen Zhang, 2025-ongoing. Phosphate-based self-sensing and energy storage cementitious materials (Co-supervisions with Prof. Lizhong Wang)
2. Changrui Dong, 2023-ongoing. Research on Bio-grouting technology for road foundation reinforcement based on water glass regulation (Co-supervisions with Prof. Zhen Guo)
3. Yongqiang Zhu, 2023-ongoing. Research on polymer-regulated microbial grouting for coastal erosion control (Co-supervisions with Prof. Zhen Guo)
4. Hengyu Liu, 2023-ongoing. Submarine Pipeline Flow Solidification Soil Protection Technology and Mechanism (Co-supervisions with Prof. Lizhong Wang)
5. Beifeng Lv, 2023-ongoing. Airport runway geopolymer bonding technology and in-situ evaluation (Co-supervisions with Prof. Lizhong Wang)

Phil&M.Sc.

Zhejiang University

1. Yichen Cao, 2025-ongoing. Study on interfacial bonding properties of phosphate-based cementitious materials (Co-supervisions with Prof. Zhen Guo)

2. Yihan Liu, 2024-ongoing. Study on the horizontal bearing capacity of monopile foundation reinforced with mixed microorganisms: from 1-g to high-g (Co-supervisions with Prof. Zhen Guo)
3. Zichun Wang, 2023-ongoing. Study on in-situ reinforcement of roadbed by bio-grouting (Co-supervisions with Prof. Lizhong Wang)
4. Peng Yue, 2025. Study on scour test and solidified soil protection of offshore wind turbine bucket foundation (Co-supervisions with Prof. Zhen Guo)
5. Minghui Fang, 2024. Experimental study on the stability of ancient seawall and deformation prediction methods (Co-supervisions with Prof. Zhen Guo)
6. Yilong Li, 2022. Experimental study on MICP reinforcement of sandy slope soil (Co-supervisions with Prof. Zhen Guo)
7. Xin Hua, 2022. Research on scour and protection of mono pile foundation under the combined action of waves and currents (Co-supervisions with Prof. Zhen Guo)

Reviewing

1. Reviewed papers for Engineering Geology, Computers and Geotechnics, Canadian Geotechnical Journal, Ocean Engineering, Applied Ocean Research, Acta Geotechnica, Biogeotechnics, Journal of Materials in Civil Engineering-ASCE, Journal of Rock Mechanics and Geotechnical Engineering, Construction and Building Materials, Powder Technology, Transportation Geotechnics
2. Reviewed proposals for National Agency for Research and Development (ANID)

Organizing/Scientific Committee

2026	Member of the Academic Committee of The 8th Youth Academic Forum on Environmental Geotechnical Engineering
2024	Member of the Organizing Committee of the West Lake Forum on Geotechnical Engineering
2024	Member of the Organizing Committee of the 2024 Marine Civil Engineering Technology Innovation Conference
2022	Member of the Organizing Committee of the 5th National Symposium on Geotechnical Engineering of Marine

Oral presentations at academic conferences

1. Microbial Optimization and Mineralization Applications in Marine Environments, the 9th National Symposium on Island Geotechnical Engineering, Xiamen, China, May, 2026.
2. Microbial mineralization rate control technology, the 9th China Hydraulic Rock Mechanics Conference, Chengdu, China, May, 2025.
3. Microbial in-situ self-repair technology of pavement, the 11th Science and Technology Innovation Progress Forum, Beijing, China, Dec, 2024.
4. Experiments of Beach Microbiological Reinforcement, the 3rd National Conference on Microbiological Geotechnical and Material Engineering, Xiamen, China. May, 2023.
5. Field implementations to resist the coastal erosion of the sandy slope by eco-friendly methods, Sino-UK Forum on Offshore Resilience Engineering Technology, Qingdao, China, July, 2023 (Invited).
6. Multi-process reinforcement of sand by MICP, the 5th National Symposium on Island Geotechnical Engineering, Deqing, China, Aug, 2022 (Invited).

Peer-reviewed publications (* represent corresponding author)

1. Zhu, Y., Guo, Z., Lu, Z*., **Li, Y*.,** Liu, Y., & Dong, C. (2026). Mechanisms of PA-Assisted MICP Reinforcement with High Concentration Few Treatments strategy: Insights from a Microfluidic Perspective. Biogeotechnics, 100248.

2. Dai, Z., Liu, H., Lv, B., Wang, L., **Li, Y*.,** & Guo, Z. (2026). Solidification of marine silt with a one-part slag-based binder: underwater performance and microscopic mechanisms. Applied Ocean Research, 174, 105198.
3. Dong, C., Guo, Z., Wang, L., & **Li, Y*.** (2026). Water Glass Modified Bioslurry for MICP under High Concentrations Calcium: Rapid Strength Development. Canadian Geotechnical Journal, (ja).
4. Dong, C., Guo, Z., Wang, L., & **Li, Y*.** (2026). Water Glass Modified Bioslurry for MICP under High Concentrations Calcium: LF-NMR Insights into High Ca 2+ Tolerance. Canadian Geotechnical Journal, (ja).
5. **Li, Y.,** Wang, L., Lv, B., Yue, P., Zhang, Z., & Guo, Z. (2026). Transformation of dredged silt into fluidized solidified soil for cross-sea bridge scour protection. Transportation Geotechnics, 102097.
6. **Li, Y.,** Dong, C., Mei, S., Lv, B., Sun, X., & Zhu, P. (2026). A coupled BCHW model for predicting wave-induced residual liquefaction of bio-grouted seabed around buried tunnels. Ocean Engineering, 356, 125255.
7. **Li, Y.,** Wang, L. Z., He, X., Mei, S., Cui, M., & Guo, Z. (2026). Coupled modeling of one-phase EICP and suction-assisted grouting for ground improvement. Canadian Geotechnical Journal, 63, 1-24.
8. **Li, Y.,** Wang, L., Wang, Z., & Guo, Z. (2026). Numerical Investigation on the Microbial Shallow Grouting Reinforcement in Permeability Anisotropic Seabeds and Liquefaction Evaluation. International Journal of Geomechanics-ASCE, 26(6), 04026107.
9. Zhu, Y., Yue, P., **Li, Y.,** Zhang, Z., & Guo, Z. (2026). Durability of PA-assisted MICP-reinforced sand in marine environment. Journal of Rock Mechanics and Geotechnical Engineering.
10. Yang, Z., Guo, Z., Liu, H., Zhang, J., Xu, W., & **Li Y***. (2026). Early strength at hour level and influencing mechanisms of geopolymer cemented granular materials: A DEM study considering realistic particle shapes. Construction and Building Materials, 509, 145237.
11. Men, S., Peng, M., Chen, H., **Li Y*.** (2026). Evaluation of drainage conditions during uplift using a new dimensionless velocity. Ocean Engineering, 2026, 251(1): 124376.
12. Dong, C., Guo, Z., Liu, H., **Li, Y.,** & Rui, S. (2026). Water glass-enhanced biocementation: Crystal precipitation evidence and environmental dependence analysis. Journal of Rock Mechanics and Geotechnical Engineering.
13. Lv, B., Wang, L., Zeng, Q., Liu, H., **Li, Y.,** & Guo, Z. (2026). Autogenous shrinkage mitigation effect of superabsorbent polymers on alkali-activated GGBFS-FA binary binder without additional water: Performance, microstructure and mechanism. Cement and Concrete Research, 199, 108058.
14. He, B., Yue, P., Zhu, Y., Liu, H., Guo, Z., & **Li, Y*.** (2025). Experimental study on hole erosion behaviors of marine soil solidified by an innovative organic composite material. Applied Ocean Research, 163, 104755.
15. **Li, Y.,** Guo, Z., Wang, L., Sun, X., & Zhu, Y. (2025). Numerical analysis of microbially induced calcite precipitation and enzyme induced calcite precipitation in calcareous sand: Multi-process and biochemical reactions. Journal of Rock Mechanics and Geotechnical Engineering, 17(5), 3225-3242.
16. **Li, Y.,** Wang, L., Dong, C., Feng, G., Sun, X., & Guo, Z. (2025). A coupled mathematical model of microbial grouting reinforced seabed considering the response of wave-induced pore pressure and its application. Computers and Geotechnics, 184, 107235.
17. Zhu, Y., **Li, Y.,** Sun, X., Guo, Z., Rui, S., & Zheng, D. (2025). A one-phase injection method to improve the strength and uniformity in MICP with polycarboxylic acid added. Acta Geotechnica, 1-13. doi.org/10.1007/s11440-024-02501-x.
18. Sun, X., Wang, J., Mao, J., Wu, L., Zeng, J., Wei, H., Pan, D., **Li, Y.,** & Guo, Z. (2025). Evolution of the soil dynamic response during the vibratory pile driving based on an integrated MPM-FEM method. Computational Particle Mechanics, 1-23.
19. Sun, X., Zhu, S., Hu, Z., Chen, W., Wei, H., Pan, D., **Li, Y.,** Fang, M., Guo, Z. (2025). Prediction of stability and deformation for seawalls using RSM-SVR method. Marine Georesources & Geotechnology, 1-24.
20. Liu, S., Dong, C., Zhu, Y., Wang, Z., **Li, Y*.,** & Feng, G. (2025). Recent Developments on Biomineralization for Erosion Control. Applied Sciences, 15(12), 6591.
21. Dong, C., Guo, Z., Zhu, Y., Rui, S., & **Li, Y*.** (2025). Effect of Water Glass-Enhanced Biocemention in Sand: Early Strength Enhancement and Mechanistic Insights. Construction and Building Materials, 479:141516.
22. **Li, Y.,** Yue, P., Zhu, Y., Sun, X., & Guo, Z. (2024). Field erosion protection of tidal beach: Comparison and synergy between sulfide aluminum cementing materials and biomineralization materials. Case Studies in

Construction Materials, 21, e04092.

23. **Li, Y.,** Guo, Z., Wang, L., & Zhu, Y. (2024). Numerical Study of MICP Infiltration and Mineralization in Unsaturated Soils: CaCO3 Distribution and Critical Depth. International Journal of Geomechanics-ASCE, 24(11), 04024258.
24. **Li, Y.,** Guo, Z., Wang, L., Zhu, Y., & Rui, S. (2024). Field implementation to resist coastal erosion of sandy slope by eco-friendly methods. Coastal Engineering, 189, 104489.
25. Yue, P., He, B., **Li, Y.,** Zhu, Y., Xu, Q., Guo, Z. (2024). Experimental investigation on local scour of bucket foundation considering exposed height and hydraulic condition. Journal of Zhejiang University (Engineering Science), 58(11):2347-2354. (In Chinese)
26. Zhu, X., Wang, J., Wang, H., & **Li, Y*.** (2024). A numerical model of the MICP multi-process considering the scale size. Plos one, 19(1), e0297195.
27. Feng, G., Xu, C., Ding, Z., Liang, L., **Li, Y.,** & Chi, M. (2024). Simplified method for evaluating tunnel response induced by a new tunnel excavation underneath. International Journal for Numerical and Analytical Methods in Geomechanics, 48(7), 1797-1813.
28. **Li, Y.,** Guo, Z., Wang, L., & Yang, H. (2023). A coupled bio-chemo-hydro-wave model and multi-stages for MICP in the seabed. Ocean Engineering, 280, 114667.
29. **Li, Y.,** Guo, Z., Yang, Z., Li, Y., Xu, M. (2023). Experimental study on the reaction process and engineering characteristics of marine calcareous sand reinforced by eco-friendly methods. Applied Ocean Research, 138, 103641.
30. **Li, Y.,** Li, Y., Guo, Z., & Xu, Q. (2023). Durability of MICP-reinforced calcareous sand in marine environments: Laboratory and field experimental study. Biogeotechnics, 1(2), 100018
31. **Li, Y.,** Rui, S., Li, L., Guo, Z., & Sun, X. (2023). A substitute for Portland cement: experiments on ecofriendly reinforcement of large-scale calcareous sand by microbial-induced carbonate precipitation spraying method. Sustainability, 16(1), 225.
32. Xu, M., Zhou, W., Shen, K., **Li, Y.,** & Guo, Z. (2023). Multi-springs model for suction caisson under monotonic lateral loading. Computers and Geotechnics, 164, 105827.
33. Feng, G., Chen, Q., Xu, C., Wan, P., Sun, F., **Li, Y.,** & Sun, Z. (2023). Improved theoretical solutions for estimating the tunnel response induced by overlying excavation. Sustainability, 15(3), 2589.
34. **Li, Y.,** Guo, Z., Li, Y., Rui, S., & Zhu, Y. (2023). Advances in MICP reinforcement technology used in island engineering. Chinese Journal of Engineering, 45(5), 819-832. (In Chinese)
35. Li, Y., Guo, Z., Xu, Q., & **Li, Y.** (2022). Experimental research on wet-dry cycle of MICP cemented calcareous sand in seawater environment. J. Zhejiang Univ. (Eng. Sci.), 56(09), 1740-1749. (In Chinese)
36. **Li, Y.,** Guo, Z., Wang, L., Yang, H., Li, Y., & Zhu, J. (2022). An innovative eco-friendly method for scour protection around monopile foundation. Applied Ocean Research, 123, 103177.
37. Hua, X., **Li, Y.,** Liu, H., Guo, Z. (2022). Experimental study on scour protection of MICP around monopile in waves and current action. Journal of Ground Improvement,4(03):190-200. (In Chinese)
38. Fang, M., **Li, Y.,** Shen, K., Wang, K., Guo Z. (2022). Experimental study on calcareous sand-steel interface shearing considering the effects of the normal stiffness. Journal of Ground Improvement, 4(01):17-24. (In Chinese)
39. Li, Y., Xu, Q., **Li, Y.,** Li, Y., & Liu, C. (2022). Application of microbial-induced calcium carbonate precipitation in wave erosion protection of the sandy slope: An experimental study. Sustainability, 14(20), 12965.
40. Baofeng Z, **Yujie L,** Huafeng Y, et al. Mechanical Characteristics of MICP-Treated Calcareous Sand and Natural Coral Rock[C]//ISOPE International Ocean and Polar Engineering Conference. ISOPE, 2022: ISOPE-I-22-471.
41. Yang, H., Guo, Z., **Li, Y.,** Wang, H., & Dou, Y. (2021). Experimental study on microbubble drag reduction on soil-steel interface. Applied Ocean Research, 116, 102891.
42. **Li, Y.,** Guo, Z., Wang, L., Ye, Z., Shen, C., & Zhou, W. (2021). Interface shear behavior between MICP-treated calcareous sand and steel. Journal of Materials in Civil Engineering-ASCE, 33(2), 04020455.
43. Zhou, W., Guo, Z., Wang, L., Li, J., **Li, Y.,** & Rui, S. (2021). Simplified t-z models for estimating the frequency and inclination of jacket supported offshore wind turbines. Computers and Geotechnics, 132, 103959.
44. Rui, S., Wang, L., Guo, Z., Zhou, W., & **Li, Y.** (2021). Cyclic behavior of interface shear between carbonate

sand and steel. Acta Geotechnica, 16, 189-209.

45. **Li, Y.,** Guo, Z., Wang, L., Li, Y., & Liu, Z. (2020). Shear resistance of MICP cementing material at the interface between calcareous sand and steel. Materials letters, 274, 128009.
46. Rui, S., Guo, Z., Si, T., & **Li, Y.** (2020). Effect of particle shape on the liquefaction resistance of calcareous sands. Soil Dynamics and Earthquake Engineering, 137, 106302.

Patents and software

1. **Li, Y.,** Guo, Z., Rui, S., Zhao, S., Zhou, W., Zhu, C. A device and technology capable of realizing MICP grouting for strengthening soil and removing by-products simultaneously, Chinese Patent, (Patent No CN110185024 B, grant data 22.05.2020).
2. **Li, Y.,** Guo, Z., Xu, M., Fang, M., Zhu, Y. A test device and method for microbial mineralization reinforcement of unsaturated soil, Chinese Patent, (Patent No CN115326667B, grant data 01.08.2023).
3. **Li, Y.,** Guo, Z., Zhu, Y., Yue, X. Double liquid spraying reinforcement device for island and reef shore slope, Chinese Patent, (Patent No CN 117623682 B, grant data 16.12.2025).
4. **Li, Y.,** Guo, Z., Lai, Y., Yang, Z., Yue, P. A large-scale sand ecological reinforcement monitoring and evaluation test system simulating marine environment, Chinese Patent, (Patent No CN116256494 A, grant data 18.06.2024).
5. Zhu, Y., **Li, Y.,** Guo, Z., Zheng, D., Wang, L. A ground reinforcement solution, its preparation method and application, Chinese Patent, (Patent No CN117623682 B, grant data 16.12.2025).
6. Wang, L., Guo, Z., **Li, Y.,** Li, L., Hong, Y., Li, J., Rui, S. SELF-DRIVING SENSOR CAPABLE OF BEING USED FOR TESTING IN REAL TIME LATERAL RESISTANCE AND END RESISTANCE OF MARINE PIPE PILE, AND TESTING DEVICE, Japanese Patent, (Patent No 2020-079309, grant data 13.10.2020).
7. Guo, Z., Wang, L., **Li, Y.,** Hong, Y., Li, L., Zhou, W., Rui, S. VISUAL INTERFACE-BASED DIRECT SHEAR APPARATUS CAPABLE OF TAKING TEMPERATURE AND SEEPAGE EFFECT INTO CONSIDERATION, Japanese Patent, (Patent No 2020-557981, grant data 22.09.2021).
8. Rui, S., **Li, Y.,** Guo, Z., Wang, L., Zhou, W., Zhu, C., Zhu, Z. A suction bucket foundation capable of extending side plates and bearing higher pull-up load, Chinese Patent, (Patent No CN 110344424 B, grant data 22.05.2020).
9. Rui, S., **Li, Y.,** Guo, Z., Zhou, W., Wang, L. Test method for tangential and normal resistance of anchor chain and soil under equivalent elastic boundary, Chinese Patent, (Patent No CN 109374418 B, grant data 12.05.2020).
10. Rui, S., **Li, Y.,** Zhou, W., Guo, Z., Zhao, S., Wang, L. A high-grip anchor with a retractable wing and a method for installing and recovering the anchor, Chinese Patent, (Patent No CN 109178210 B, grant data 16.07.2019).
11. Rui, S., **Li, Y.,** Guo, Z., Wang, L., Zhou, W., Zhu, C., Zhu, Z. A new type of bucket foundation with expandable mud panels, Chinese Patent, (Patent No CN 110374129 B, grant data 14.07.2020).
12. **Li, Y.,** Guo, Z., Dou, Y., et al. A kind of beach MICP reinforcement analysis software considering Marine dynamic environment V1.0, Chinese Software Copyright, (No. 2022SR0127043).
13. **Li, Y.,** Guo, Z., Hua, X., et al. A dynamic theoretical model of microbial mineralization deposition and a site reinforcement analysis software V1.0, Chinese Software Copyright, (No. 2022SR0127042).
14. **Li, Y.,** Guo, Z., Wang, L., et al. A utility model relates to an analysis software for mineralization reinforcement process of microbial unsaturated infiltration V1.0, Chinese Software Copyright, (No. 2023SR0666764).
15. **Li, Y.,** Guo, Z., Wang, L., et al. An invention relates to a kinetic theoretical model of enzyme-promoted calcium carbonate deposition and a foundation consolidation analysis software V1.0, Chinese Software Copyright, (No. 2023SR0666765).